

From the Warehouse Map to a Saved Audit

How Inspectors Scan SKUs on a Rack

A walkthrough of the tablet inspector flow: finding a task on the warehouse map, opening a rack's canvas, running a live SKU scanning session, and how each outcome (matched, mismatched, missing) gets recorded and saved — including every distinct use case the flow supports.

App

Rack Audit — React Native / Expo (tablet, inspector-facing)

Branch

variation-2

Scope

Warehouse Map → Rack View → Live Scan
→ Save

Audience

Internal R&D review

1 End-to-End Flow at a Glance

An inspector starts from a list of assigned tasks, locates the physical rack visually on a warehouse-wide map, opens that rack's scanning canvas, and works through it pallet by pallet with the tablet camera — every scan is saved the moment it resolves, with no separate "submit" step at the end.



2 Step 1 — Finding the Rack: Warehouse Map

From "My Audit Tasks," tapping **View Tasks on 3D** opens a top-down visual layout of the warehouse (Layout → Rack → Bay), rather than a text list — useful for judging physical distance/effort before committing to a task.

- Every rack in the warehouse is shown as a card with a row of small colored bay segments, grouped by which layout it belongs to.
- Rack color reflects status at a glance: ● green = fully **Completed**, ● light blue = **Assigned** (has a task, not overdue), ● red = **Delayed** (overdue).
- Filters (Due Date, Task Type — Full / Cycle Count / Spot Check) narrow what's shown; overdue racks are hidden by default unless explicitly filtered in.
- Tapping a rack opens a centered detail popup listing every task touching it, grouped by layout, with a **Start Task** button that jumps straight into that rack's canvas.

3 Step 2 — The Rack Canvas (Rack View)

Rack View shows the physical rack as a schematic front elevation — every bay the rack has, side by side, not one bay at a time.

- Bays are separated by a vertical "upright" divider line, mirroring a real rack frame; level labels (L1, L2, ...) only print once, on the leftmost bay.
- Each small rectangle is one pallet slot. Pinch-to-zoom and drag-to-pan work across the whole canvas.
- **Pallet highlighting** depends on what the audit is scoped to:
 - *Regular audit (Full / Cycle Count, no target SKU)*: any pallet with an assigned SKU is highlighted a darker gray — everything relevant to scan.
 - *Spot-check audit with a target SKU* (set on the admin's Create Audit form, e.g. only check iPhone boxes / SKU-1001): only pallets actually carrying that SKU are highlighted and selectable; every other pallet is faded and disabled — you physically cannot tap or scan into it.

4 Step 3 — Starting a Live Scan Session

Unlike an older single-pallet-at-a-time flow, **Start Audit** requires no pre-selection — tap it (footer button, or the camera icon in the toolbar at any time) and a scanning overlay opens on top of the canvas.

- The canvas remains the persistent base screen underneath, dimmed but not unmounted — closing the overlay (**Done**) simply reveals it again, with all progress colors intact.
- The overlay has a live camera view up top (continuous scanning — no need to close/reopen between codes) and a running list of every SKU scanned this session below it.
- A "Use test scan" fallback simulates a scan for demoing without a physically printed code.

5 Step 4 — Scanning Logic & Every Use Case

Each scan claims the next not-yet-visited in-scope pallet on the rack (in canvas order) — the inspector is standing at a real physical pallet when they scan it, whether it turns out right or wrong, so every scan always has a real location behind it.

CASE A · MATCHED

Scanned SKU is exactly what's expected at that pallet

Pallet turns ● **green** on the canvas; list shows a "Matched" badge with the SKU, name, and real Bay/Level/Pallet location. Saved immediately — no separate confirm step.

CASE B · MISMATCH

Scanned SKU is not what's expected at that pallet

Pallet turns ● **amber/orange** on the canvas; list shows a "Mismatch" badge, still with a real location attached (not "unresolved") — the wrong item was found at a specific, identifiable pallet.

CASE C · MISSING

A pallet is in scope but hasn't been (or can't be) scanned

Shown with a ● **dashed border** on the canvas — either "not reached yet" (transient, clears once scanned) or genuinely empty/no SKU present (permanent for that session, since there's nothing to scan). Persists even after closing the Live Scan overlay, so progress is still readable on the base canvas.

CASE D · RAISE ISSUE

SKU matched, but qty or condition looks wrong

On a Matched list row, tapping the flag icon expands an inline accordion (quantity + condition fields, reusing the same card Count Sheet uses). Saving it: persists the correction, marks the row "Issue Raised," and makes that pallet **blink** on the canvas for a few seconds so it's obvious which physical pallet needs attention — a small red dot marker stays on it afterward.

CASE E · DUPLICATE-SCAN PROTECTION

Same physical label held in view too long, vs. a genuinely new one

The camera won't double-count a lingering code (holding on one pallet's label doesn't register two scans), but a different code appearing — even mid-pan across a dense sheet of codes — registers immediately, without waiting for a cooldown.

6 Step 5 — Saving & Where It Goes

- Every Matched or Mismatch scan is saved the instant it resolves — no batch "submit" at the end of the session.
- The first save on a location flips its status from **Not Started** to **In Progress**; once every location in scope is covered, the rack/bay reads as **Completed** — which is exactly what feeds the Warehouse Map's green/blue/red coloring described in Step 1.
- A Raise Issue save persists the corrected line the same way, on top of the original match — flagged (non-Good condition) lines are what later surface in the **Reported Audits / Progress** views elsewhere in the app.
- **Done** just closes the scanning overlay — no navigation away, no data loss; a session can be reopened any time via the toolbar's camera icon and picks up exactly where it left off (already-claimed pallets stay claimed).

7 Reference — Canvas States

Visual	Meaning	Selectable?
● Green fill	Matched — scanned SKU equals what's expected	—
● Amber/orange fill	Mismatch — wrong SKU found at this pallet	—
Dashed border	Missing — in scope, not yet scanned (or empty)	—
Dark gray fill	In scope for this audit, not yet scanned	Yes
Faint / 45% opacity	Out of scope (only when a target SKU is set)	No — disabled
Small red dot	An issue was raised on this pallet	—
Blinking	An issue was just saved on this pallet (transient, ~2.5s)	—

8 Dev/Test Aid — Test Scan Sheet

For exercising this flow without physical inventory, an in-app "Test Scan Sheet" (reachable from Rack View's header menu when applicable) shows a printable page of QR codes covering all three outcomes on one sheet — a set that resolve Matched, a set that resolve Mismatch, and a set intentionally left blank to represent Missing/empty pallets — so the whole scanning flow can be validated end-to-end with a second device's camera.